SMART INDIA HACKATHON 2025
INTRODUCTION

- **Problem Statement ID – 25030**
- **Problem Statement Title- AI-BasedCrop Recommendation for Farmers**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID- 73834**
- **Team Name - Infinite Looper.**

AI-BasedCrop Recommendation for Farmers

- ## Proposed Solution

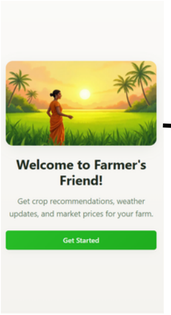
Overview of the Solution

AI-driven mobile app providing personalized crop recommendations and yield predictions through soil, weather, and market data analysis with multilingual support and ML functionality.

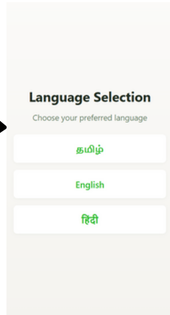
Key Features

- **AI-Powered Recommendations**: Analyzes soil, weather, and market data to recommend the most suitable crops.
- **Predictive Analytics**: Forecasts yield, profit, and sustainability scores.
- **Conversational Interface**: A multilingual voice and chat feature provides actionable responses to farmer queries.
- **Real-Time Data Integration:** Pulls data from satellite APIs (e.g., SoilGrids), weather services, and market price trackers.
- **Offline Functionality:** The app works in low-connectivity regions with on-device models for tasks like image-based disease detection.
- **User-Friendly Design:** A simple, intuitive interface tailored for all skill levels

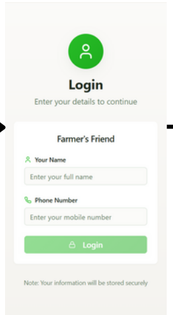
Click for Demo video: Demo video Link



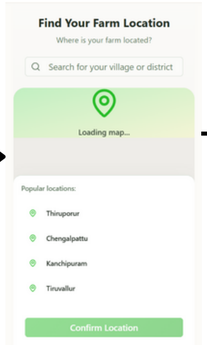
Welcome page



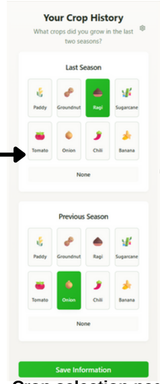
Language selection page



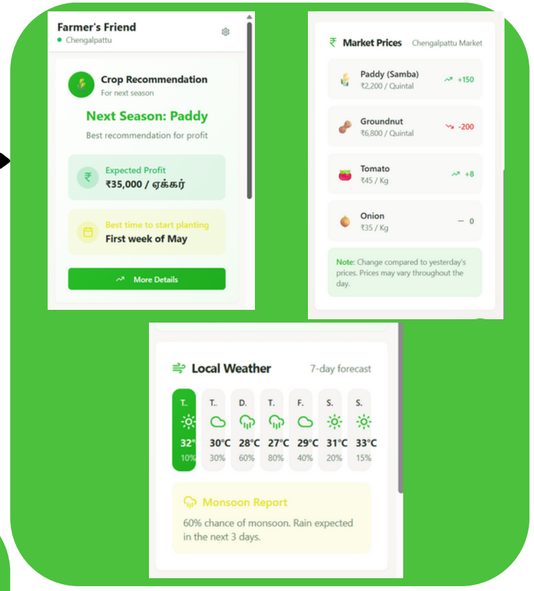
Login page



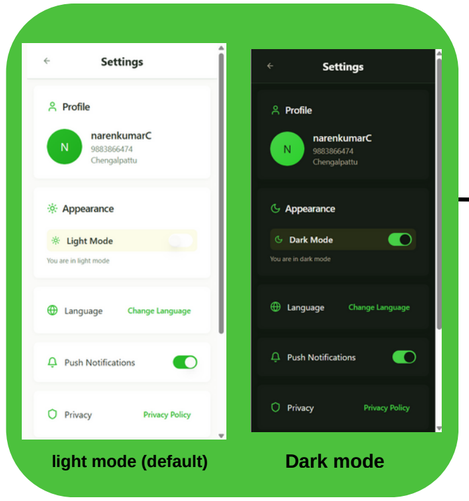
Location of the user



Crop selection page



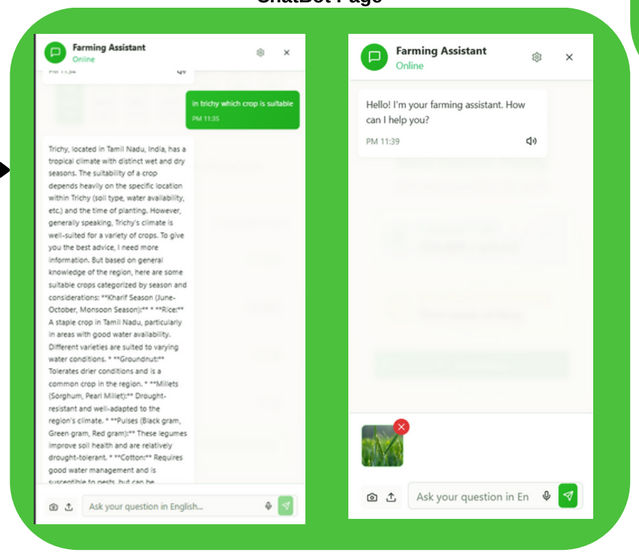
Home page



light mode (default) Dark mode

Settings

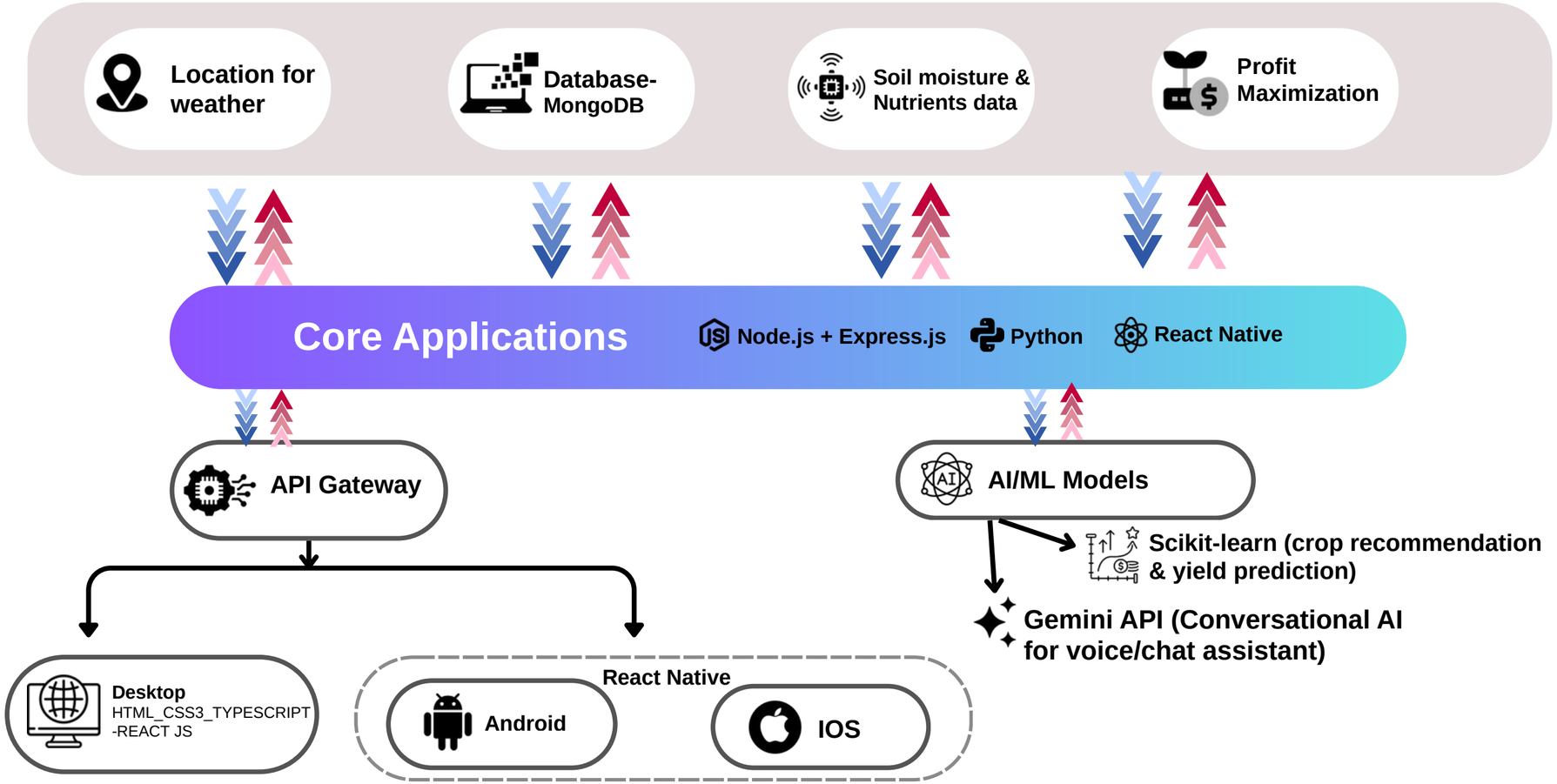
ChatBot Page



Wire frame

TECHNICAL APPROACH

TECHNOLOGY STACK



Technical Approach & Methodology

Phase 1: Foundation & Data Layer

- Set up Node.js backend, MongoDB, and data pipelines (weather, soil, market).

Phase 2: Core AI Integration

- Build Python ML models (crop, yield) with Flask/FastAPI.
- Connect ML services to backend for predictions.

Phase 3: Mobile App Development

- Develop React Native app (Android/iOS).
- Integrate APIs, TensorFlow Lite (offline disease detection), and Gemini AI chatbot.

Phase 4: Deployment & Scaling

- Deploy on AWS/GCP.
- Conduct farmer field testing and refine.

FEASIBILITY

HIGH ACCURACY

- Random Forest (99%)
- Gradient Boosting (99.27%) confirm technical feasibility.

DATA AVAILABILITY

- Open Kaggle dataset with soil, weather, and pH.

EFFICIENCY & INTERPRETABILITY

- Models train on normal systems, no GPU needed.
- Feature importance (Weather) supports Explainable AI.

SCALABILITY

- Models can be saved, deployed on apps, IoT, or cloud.

VIABILITY

RELEVANCE

- Helps farmers choose crops under climate uncertainty.

OPTIMIZATION

- Guides efficient irrigation and fertilizer use.

SUSTAINABILITY

- Supports precision agriculture and UN SDGs.

ADOPTION

- 99% accuracy boosts farmer trust, esp. with IoT/weather API integration.

INDEPENDENCE

- Works across crops and locations.

IMPACT AND BENEFITS

INFINITE LOOPERS.

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BENEFITS

- **Personalized Recommendations:**
  Tailored to the farmer's soil and climate conditions.
- **Improved Yield:**
  Right crop choice improves productivity and profitability.
- **Resource Optimization:**
  Helps farmers use water, fertilizer, and pesticides efficiently.
- **Accessibility:**
  Overcomes language and knowledge barriers with AI-driven natural language support.
- **Disease & Risk Reduction:**
  AI can predict risks (e.g., drought impact, pest risk) before they occur.

IMPACTS

- **Economic Growth:**
  Increased crop yield boosts farmer income and reduces crop failure losses.
- **Sustainability:**
  Encourages smart farming, reducing overuse of fertilizers and conserving water.
- **Social Impact:**
  Empowers small and marginal farmers who lack access to expert advice.
- **Digital Inclusion:**
  Promotes AI adoption in rural communities, bridging the technology divide.

BENEFITS

Increased Income & Reduced Costs

Empowered Decision-Making

Accessible Knowledge

Sustainable Farming

Risk Reduction

Reduced Manual Labor/Time Savings

IMPACTS

Economic Growth
Increased crop yield boosts farmer income and reduces crop failure losses.

Digital Inclusion
Promotes AI adoption in rural communities, bridging the technology divide.

Sustainability
Encourages smart farming, reducing overuse of fertilizers and conserving water.

Social Impact
Empowers small and marginal farmers who lack access to expert advice.

RESEARCH AND REFERENCES

IMPROVEMENTS (Based on ScienceDirect Papers):

- **Uncertainty Quantification (UQ)**: Unlike most models, the paper added entropy-based UQ — future work should explore more robust UQ techniques to increase trustworthiness in real-world use. [1]
- **Sensor & IoT Integration:** The paper highlights that incorporating real-time soil and weather sensor data can make crop recommendations adaptive and context-specific.[2]
- **Geographic Generalization**: Current dataset is India-focused. Improvement involves expanding to diverse agro-climatic regions for broader applicability.[3]
- **Noise Robustness**: Testing the model under noisy or incomplete data ensures stable recommendations even in imperfect field conditions.[4]

REFERENCES:

- Shastri, S., Kumar, S., Mansotra, V. et al. Advancing crop recommendation system with supervised machine learning and explainable artificial intelligence. Sci Rep 15, 25498 (2025). https://doi.org/10.1038/s41598-025-07003-8
- https://www.nature.com/articles/s41598-025-07003-8?utm_source=chatgpt.com
- https://www.mdpi.com/2078-2489/16/8/632?utm_source=chatgpt.com
- An approach for crop recommendation with uncertainty quantification based on machine learning for sustainable agricultural decision-making
- Author links open overlay panelMd. Sakib Bin Alam [a], Vatcharaporn Esichaikul [b], Aiman Lameesa [c], Shams Forruque Ahmed [d e], Amir H. Gandomi [f g h]